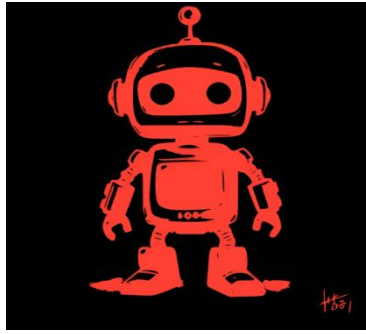




AEGIS

The Alchemists

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USER MANUAL

COS301 AEGIS TEAM
UNIVERSITY OF PRETORIA
PRETORIA 012

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AEGIS is an advanced technical assessment platform developed by The Alchemists, the COS301 AEGIS Team at the University of Pretoria. The platform combines technical challenges with integrity analytics to support the evaluation of genuine human reasoning.....	2
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INTRODUCTION

AEGIS is an advanced technical assessment platform developed by The Alchemists, the COS301 AEGIS Team at the University of Pretoria. The platform combines technical challenges with integrity analytics to support the evaluation of genuine human reasoning.

USER GROUPS

RECRUITERS:

Recruiters create and manage assessments, configure question banks, assign tests, and review assessment and integrity information.

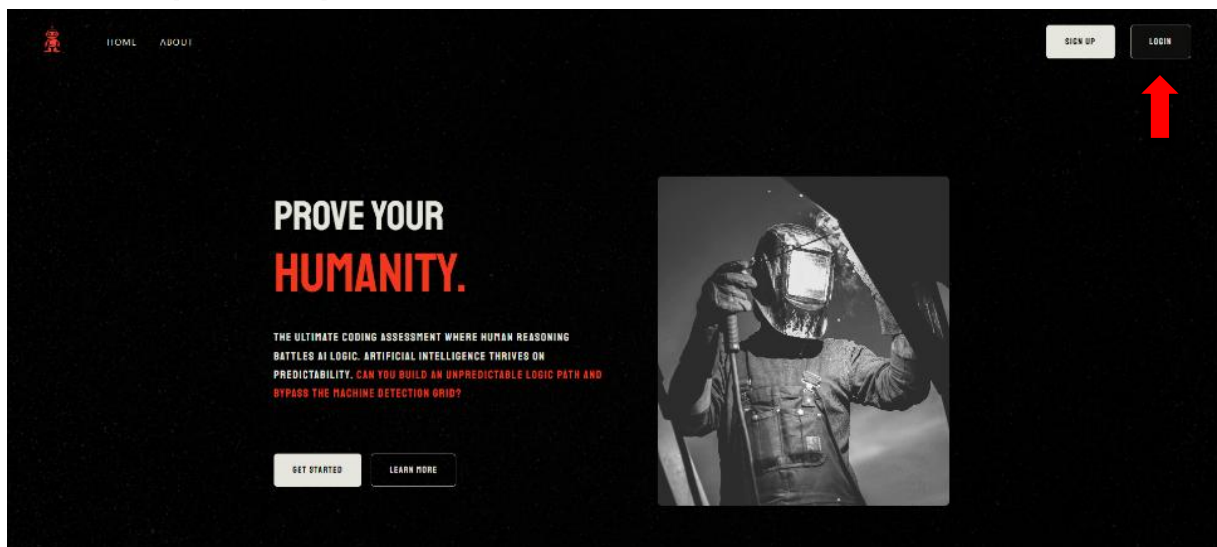
CANDIDATES:

Candidates complete assigned questions under timed conditions to demonstrate their understanding and problem-solving ability.

GETTING STARTED:

LANDING PAGE

The landing page introduces AEGIS and provides the main entry point to the application. Use the navigation or Sign In action to continue.



SIGNING IN

Open the Sign In page.

Enter the email address associated with your account.

Enter your password.

Select Sign In to submit your credentials.

WELCOME BACK

Sign in with

 **GOOGLE**

OR

EMAIL
Enter your email

PASSWORD
Enter your password 

SIGN IN

Don't have an account? [Sign Up Now.](#)

RECRUITER WORKFLOW

Recruiters use the application navigation to manage questions, create assessments, and review candidate results.

MAIN NAVIGATION

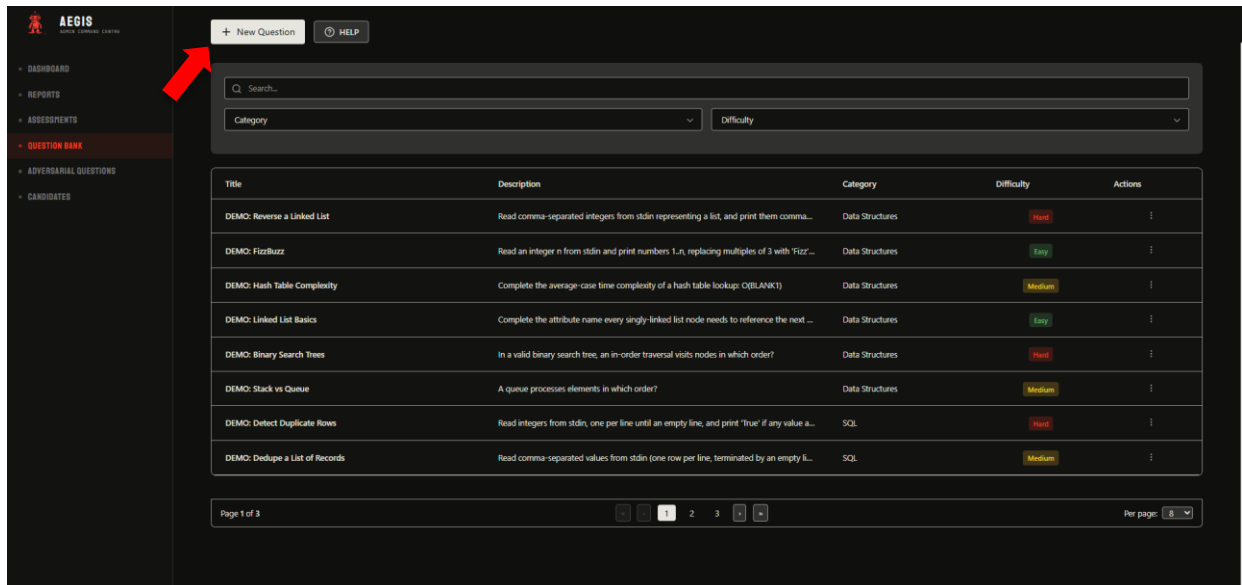
- **Dashboard:** Provides a high-level view of assessment activity and system signals.
- **Reports:** Provides assessment and candidate performance information.
- **Assessments:** The area used to manage assessments.
- **Question Bank:** The question-management area shown in the application navigation.
- **Adversarial Questions:** The area shown for adversarial-question content.
- **Candidates:** The area shown in the application navigation.

QUESTION BANK

The Question Bank is the central repository for standard technical questions. The supplied interface supports question types including coding, multiple-choice, and fill-in-the-blank questions.

STANDARD QUESTION BANK

Serves as the central repository for standard technical questions across domains such as coding, MCQ, and fill-in-the-blank questions.



The screenshot displays the AEGIS Question Bank interface. A red arrow points to the '+ New Question' button in the top navigation bar. The interface includes a sidebar with navigation links, a search bar, and a table of questions.

Title	Description	Category	Difficulty	Actions
DEMO: Reverse a Linked List	Read comma-separated integers from stdin representing a list, and print them comma...	Data Structures	Hard	
DEMO: FizzBuzz	Read an integer n from stdin and print numbers 1..n, replacing multiples of 3 with 'Fizz'...	Data Structures	Easy	
DEMO: Hash Table Complexity	Complete the average-case time complexity of a hash table lookup: O(BLANK1)	Data Structures	Medium	
DEMO: Linked List Basics	Complete the attribute name every singly-linked list node needs to reference the next ...	Data Structures	Easy	
DEMO: Binary Search Trees	In a valid binary search tree, an in-order traversal visits nodes in which order?	Data Structures	Hard	
DEMO: Stack vs Queue	A queue processes elements in which order?	Data Structures	Medium	
DEMO: Detect Duplicate Rows	Read integers from stdin, one per line until an empty line, and print 'True' if any value a...	SQL	Hard	
DEMO: Dedupe a List of Records	Read comma-separated values from stdin (one row per line, terminated by an empty li...	SQL	Medium	

Page 1 of 3

Per page: 8

Create a question: Select New Question.

Choose the content type: Select the appropriate question type and complete the required fields.

STANDARD QUESTION CREATION

A standard question should contain the information required for a candidate to understand the task and the tests required for automated evaluation.

The screenshot shows the 'CREATE QUESTION' form with the 'QUESTION METADATA' section expanded. It includes fields for 'QUESTION TITLE' (e.g., ZigZag Conversion), 'Description' (Describe the constraints, requirements and edge cases here...), 'CATEGORY' (Select Category...), 'DIFFICULTY' (EASY, MEDIUM, HARD), 'SCORE ALLOCATION' (10), and 'INDEXED TAGS' (Type a tag). The 'Save' button is highlighted in red.

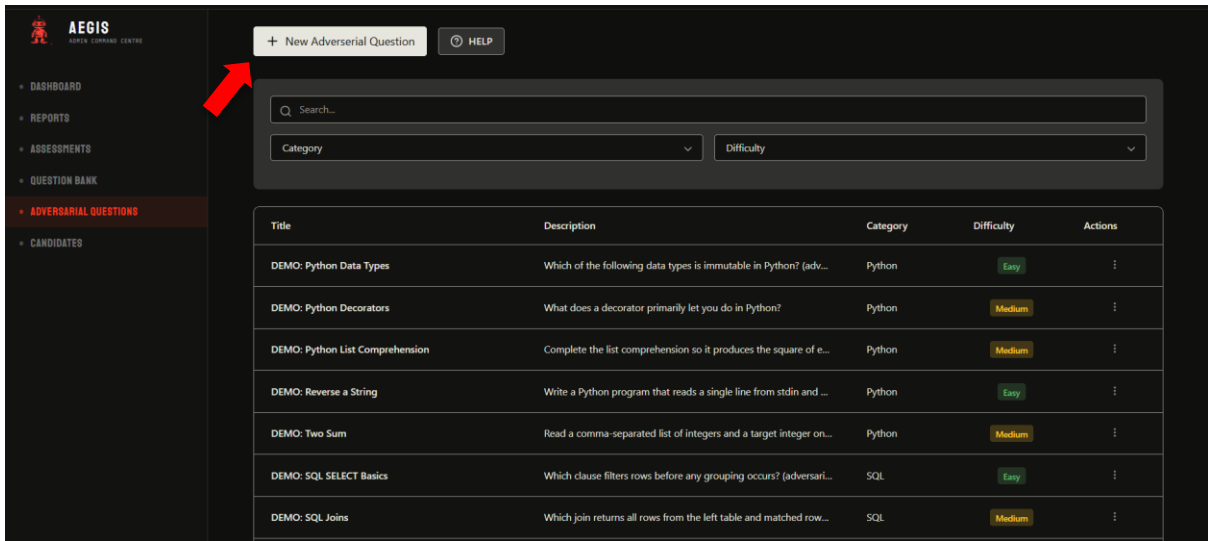
1. Question title and detailed problem prompt
2. Input and output constraints
3. Sample test cases
4. Hidden unit tests for automated grading

5.

The screenshot shows the 'CREATE QUESTION' form with the 'SOLUTION ARCHITECTURE' and 'TEST CASES' sections expanded. The 'SOLUTION ARCHITECTURE' section includes a 'FUNCTION SIGNATURE' field (def two_sum(nums, target)) and a 'PYTHON' code editor. The 'TEST CASES' section includes a 'PYTHON LITERAL INPUT' field ([(1, 2), 3]). The 'Save' button is highlighted in red and pointed to by a red arrow.

ADVERSARIAL QUESTION CREATION

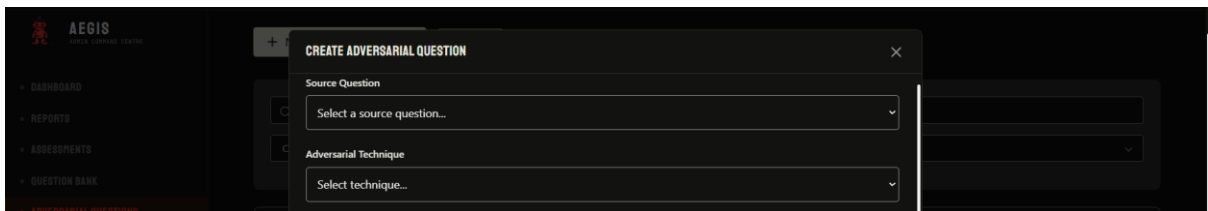
Adversarial questions are generated from an existing source question using a selected adversarial technique.



The interface shows a sidebar with navigation links: DASHBOARD, REPORTS, ASSESSMENTS, QUESTION BANK, ADVERSARIAL QUESTIONS (highlighted), and CANDIDATES. The main area features a '+ New Adversarial Question' button and a 'HELP' button. Below these is a search bar and two dropdown menus for 'Category' and 'Difficulty'. A table lists existing questions:

Title	Description	Category	Difficulty	Actions
DEMO: Python Data Types	Which of the following data types is immutable in Python? (adv...	Python	Easy	⋮
DEMO: Python Decorators	What does a decorator primarily let you do in Python?	Python	Medium	⋮
DEMO: Python List Comprehension	Complete the list comprehension so it produces the square of e...	Python	Medium	⋮
DEMO: Reverse a String	Write a Python program that reads a single line from stdin and ...	Python	Easy	⋮
DEMO: Two Sum	Read a comma-separated list of integers and a target integer on...	Python	Medium	⋮
DEMO: SQL SELECT Basics	Which clause filters rows before any grouping occurs? (adversari...	SQL	Easy	⋮
DEMO: SQL Joins	Which join returns all rows from the left table and matched row...	SQL	Medium	⋮

- Select a new adversarial question.
- Select the source question.
- Select the adversarial technique to apply.
- Generate the adversarial question.
- Review the generated question against the source question.



The modal titled 'CREATE ADVERSARIAL QUESTION' contains two dropdown menus: 'Source Question' with the placeholder 'Select a source question...' and 'Adversarial Technique' with the placeholder 'Select technique...'.

If the generated question is not suitable, use Regenerate to produce another version.

The interface is a dark-themed web application for generating adversarial questions. It features four main sections arranged in a 2x2 grid:

- ORIGINAL SOURCE QUESTION**: A text input area with the placeholder "Select source question". A red arrow points to the "GENERATE ADVERSARIAL QUESTION" button above it.
- ADVERSARIAL QUESTION**: A text output area with the placeholder "Generated adversarial question will appear here". A "REGENERATE" button is located above this section.
- ANSWER TO SOURCE QUESTION**: A text output area with the placeholder "Validation results will appear here".
- GEMINI RESPONSE TO ADVERSARIAL QUESTION**: A text output area with the placeholder "Validation results will appear here".

Navigation and Action Buttons:

- GENERATE ADVERSARIAL QUESTION**: A red button with a red arrow pointing to it.
- REGENERATE**: A light gray button located above the "ADVERSARIAL QUESTION" section.
- VALIDATE**: A red button located to the right of the "ADVERSARIAL QUESTION" section.
- DEPLOY QUESTION**: A red button with a red arrow pointing to it, located at the bottom right.

Use the trap-effectiveness information to compare generated options and support selection of the question to deploy.

TRAP EFFECTIVENESS						
observed outcomes from completed attempts						
SUGGESTED · ADVISORY SUFFICIENT DATA IRRELEVANT_CONTEXT Consider this trap for your next question — 31 completed attempts with a 48.4% review-signal rate. Highest review-signal rate among strategies with sufficient data. USE THIS TRAP						
TRAP	QUESTIONS	ATTEMPTS	ELEVATED REVIEW	REVIEW-SIGNAL RATE	EVIDENCE	
NONE_OF_THE_ABOVE ID 3	1	5	1	20.0%	SUFFICIENT DATA	SELECT
INVERTED_BOOLEAN ID 6	6	28	9	32.1%	SUFFICIENT DATA	SELECT
METHOD_SEMANTIC_INVERSION ID 4	1	7	2	28.6%	SUFFICIENT DATA	SELECT
GARDEN_PATH_CRT ID 2	3	16	5	31.3%	SUFFICIENT DATA	SELECT
IRRELEVANT_CONTEXT ID 1	7	31	15	48.4%	SUFFICIENT DATA	SELECT
NEGATION_INJECTION ID 5	2	13	5	38.5%	SUFFICIENT DATA	SELECT

ASSESSMENTS

CREATE ASSESSMENT

Assessments combine selected content into a structured test for candidates. Complete the required metadata, content selection, and integrity weighting before creating the assessment.

ASSESSMENT ARSENAL			HELP	NEW ASSESSMENT
Search assessments...				
DEMO: JUNIOR PYTHON SCREEN	DEMO: ADVANCED SQL 6 DATA STRUCTURES	DEMO: FULL STACK MIXED ASSESSMENT		
2 CANDIDATES	0 CANDIDATES	2 CANDIDATES		
0% COMPLETED	0% COMPLETED	100% COMPLETED		
2024-09-22	2024-09-29	2024-09-15		
DEMO: DATA STRUCTURES DEEP DIVE	DEMO: PYTHON FUNDAMENTALS SCREENING	DEMO: SQL QUERY CHALLENGE		
5 CANDIDATES	6 CANDIDATES	5 CANDIDATES		
100% COMPLETED	83% COMPLETED	80% COMPLETED		
2024-09-01	2024-08-04	2024-08-18		
TEST	DEMO: TWO SUM ADVERSARIAL SCREENING	TWO SUM BBD ASSESSMENT		
0 CANDIDATES	4 CANDIDATES	4 CANDIDATES		
0% COMPLETED	100% COMPLETED	75% COMPLETED		
2024-09-29	2024-09-10	2024-09-10		

Define the assessment: Enter the required assessment information and metadata

The screenshot shows the 'CREATE ASSESSMENT' dialog in the 'ASSESSMENT ARSENAL' interface. The dialog is titled 'CREATE ASSESSMENT' and has a subtitle 'Create an insight or adversarial assessment'. It is divided into four steps: 1. BASIC, 2. QUESTIONS, 3. SELECTION, and 4. CONFIRM. The first step, 'BASIC', is currently active. It contains the following fields:

- NAME:** A text input field with the value 'Testbed Backdoor Algorithm System'.
- DESCRIPTION:** A text input field with the value 'Briefly describe the purpose...'.
- SELECT TYPE:** A dropdown menu with the value 'Backdoor'.
- YOUR CERT ID:** A text input field with the value '00'.

At the bottom of the dialog, it says 'Step 1/4' and has a 'CONTINUE' button.

Select content: Choose the adversarial questions that should be included in the assessment.

The screenshot shows the 'CREATE ASSESSMENT' dialog in the 'ASSESSMENT ARSENAL' interface, now at Step 2: QUESTIONS. The dialog is titled 'CREATE ASSESSMENT' and has a subtitle 'Create an insight or adversarial assessment'. It is divided into four steps: 1. BASIC, 2. QUESTIONS, 3. SELECTION, and 4. CONFIRM. The second step, 'QUESTIONS', is currently active. It contains the following elements:

- PICK QUESTIONS:** A section with a search bar and a list of questions.
- QUESTIONS:** A list of questions with checkboxes and a 'SELECT' button.
- QUESTIONS:** A list of questions with checkboxes and a 'SELECT' button.
- QUESTIONS:** A list of questions with checkboxes and a 'SELECT' button.
- QUESTIONS:** A list of questions with checkboxes and a 'SELECT' button.
- QUESTIONS:** A list of questions with checkboxes and a 'SELECT' button.

At the bottom of the dialog, it says 'Step 2/4' and has 'BACK' and 'CONTINUE' buttons.

Configure integrity weighting: Set the integrity-score weighting for each selected question.

ASSESSMENT ARSENAL

CREATE ASSESSMENT

create and configure an assessment

WEIGHTING

Set the approved integrity weight for each question. All suggestions are reference only - accepting, editing, or rejecting to adjust an explicit value and never changes the approved weight on its own.

WHICH OF THE FOLLOWING DATA TYPES IS FUNCTIONAL IN PYTHON (ANSWERABLY REMOVED)

APPROVED WEIGHT: 0.3333333333333333

AI SUGGESTED WEIGHT: 0.25

USER WEIGHT assigned due to shorter action time and lower redundancy relative to other questions.

WHAT DOES A DECORATOR PRIMARILY LET YOU DO IN PYTHON?

APPROVED WEIGHT: 0.3333333333333333

AI SUGGESTED WEIGHT: 0.25

USER WEIGHT assigned due to moderate action time and lower redundancy relative to other questions.

Step 3/4

BACK **CONTINUE**

Create the assessment: Review the configuration and select Create Assessment to complete the process.

ASSESSMENT ARSENAL

CREATE ASSESSMENT

create and configure an assessment

CONFIRM

READY TO GO

Title: Unlabeled

Role: Backend

Time: 60 min

Questions: 2 target Q

Integrity weights: 0 accepted, 0 modified, 0 rejected

Step 4/4

BACK **CREATE ASSESSMENT**

REPORTS

Reports provide assessment-level and question-level information for reviewing candidate performance.

Open an assessment: In Reports, locate the required assessment and select View.

ASSESSMENT PERFORMANCE TABLE				
ID	Assessment	Average	Top Candidate	View
1	DEMO: Python Fundamentals Screening	50.00%	Aisha Bello	<button>VIEW</button>
2	DEMO: SQL Query Challenge	47.50%	Priya Nair	<button>VIEW</button>
9	TWO SUM BBD ASSESSMENT	44.44%	sambulo zikhali	<button>VIEW</button>

Review candidate results: Use the assessment view to review the candidates and their scores.

CANDIDATE RESULTS				
Candidate Name	Total Score	Status	Integrity Risk Score	View Answers
Luyanda Ndlovu	66.67%	PASS	71 High	<button>VIEW</button>
Myron	0.00%	FAIL	55 Medium	<button>VIEW</button>
sambulo zikhali	66.67%	PASS	27 Low	<button>VIEW</button>

Review question telemetry: Use the question navigation sidebar to select a specific question and review its telemetry results.

Q1
TWO SUM MCQ
0/10

Q2
TWO SUM FILL IN THE BLANK
0/10

Q3
TWO SUM
0/10

In the standard $O(n)$ implementation of the Two Sum problem, which uses a hash map to achieve linear time complexity, what is the primary data stored in the hash map to facilitate the lookup process? A: Indices of numbers already seen B: Sorted order of the array C: All possible sums D: Only the target value Note: The algorithm typically processes an array of size N where N is the number of elements, and the hash map is initialized with a capacity of $N/2$ to minimize collisions while maintaining a load factor of 0.5, assuming a uniform distribution of integer keys across the 32-bit signed integer space. Answer with the option letter only.

REVIEW QUESTION

URGENCY RATING
56.3
MEDIUM

CONTRIBUTING FACTORS

- Question 1 (multiple-choice): copy activity was elevated on this question (3 copy event(s), 1444 characters copied).

BEHAVIORAL TELEMETRY

PASTING VS TYPING
0% PASTED

Pasted0

Typed1444

ACTIVE TIME VS FOCUS LOSS

Active Time0m 20s

Focus Loss(1 events) 0m 0s

EDITING ACTIVITY

Backspaces0

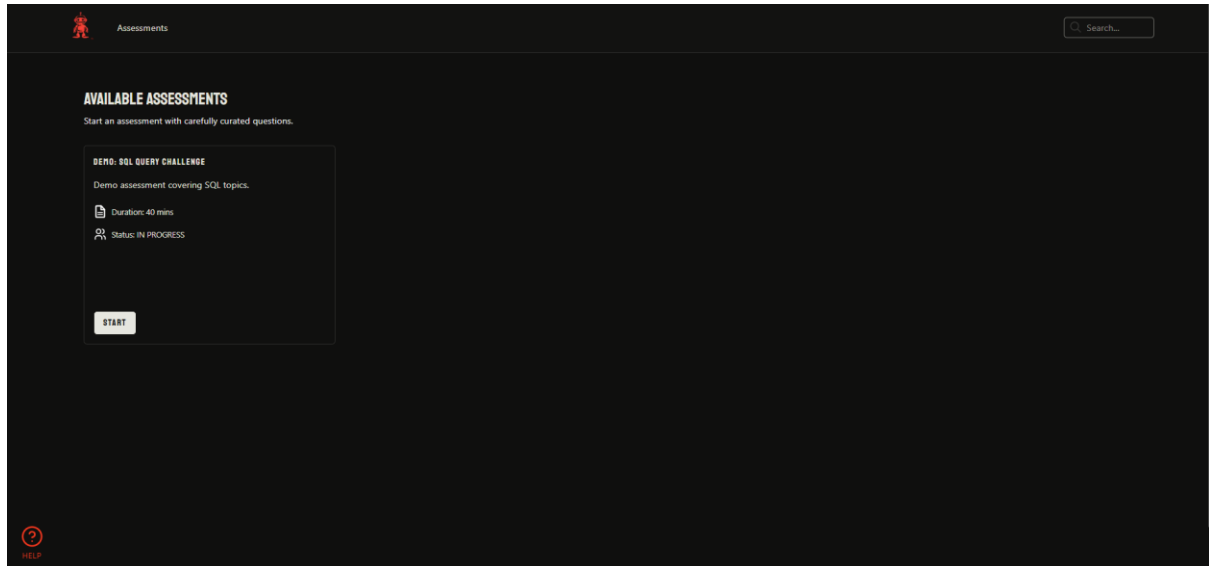
TELEMETRY LOG

99

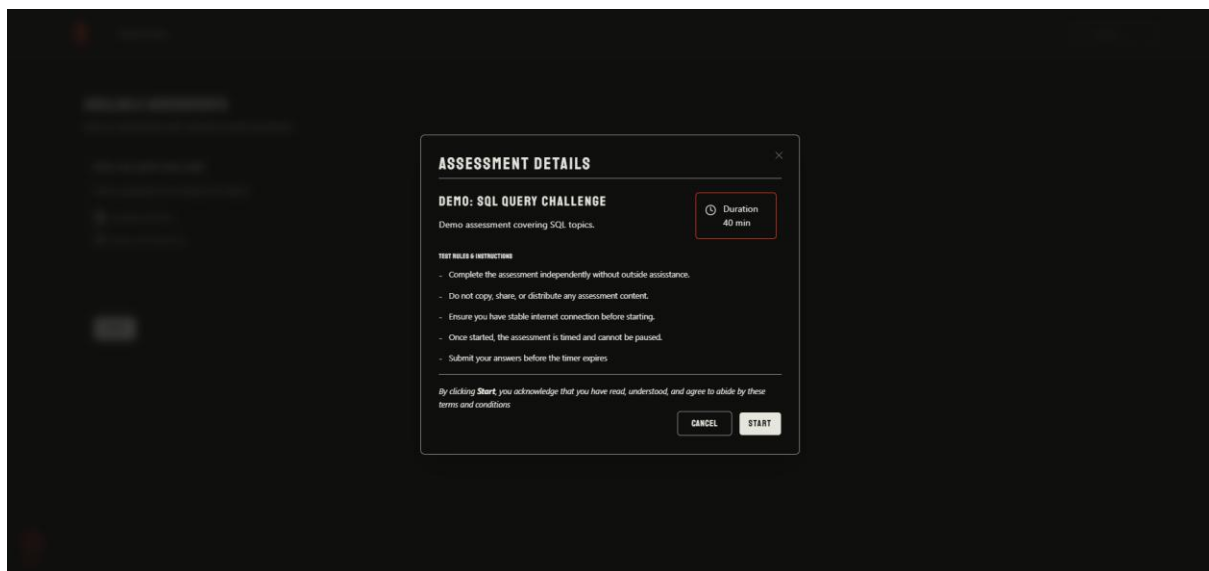
CANDIDATE WORKFLOW:

Candidates complete an assigned assessment through the assessment workspace. The workflow provides controls for starting the assessment, answering questions, running code, navigating, and submitting.

Start the assessment: Select Start to open and begin the assigned assessment.



Review the assessment: Read the assessment information and instructions before proceeding



Answer questions: Enter the required response for the current question.



TIME REMAINING: 00:59:51

EXIT SESSION

DESCRIPTION

TWO SUM MCQ

In the standard $O(n)$ implementation of the Two Sum problem, which uses a hash map to achieve linear time complexity, what is the primary data stored in the hash map to facilitate the lookup process?

A: Indices of numbers already seen
B: Sorted order of the array
C: All possible sums
D: Only the target value

Note: The algorithm typically processes an array of size N where N is the number of elements, and the hash map is initialized with a capacity of $N/2$ to minimize collisions while maintaining a load factor of 0.5, assuming a uniform distribution of integer keys across the 32-bit signed integer space. Answer with the option letter only.

MULTIPLE CHOICE

CHOOSE THE MOST CORRECT ANSWER

A. Indices of numbers already seen

B. Sorted order of the array

C. All possible sums

D. Only the target value

Run code: For coding questions, select Run Code to execute the solution against the available tests.

DESCRIPTION

TWO SUM

Implement the function `def two_sum(nums, target):` which returns the indices of the two numbers such that they add up to target. You may assume that each input would have exactly one solution, and you may not use the same element twice. The input list 'nums' may contain up to 10^6 elements, and the 'target' value is guaranteed to be an integer between -10^9 and 10^9 . For performance, the solution must run in $O(n)$ time complexity. Note that the function must return the indices in the order they appear in the original list, and the input list is guaranteed to be unsorted. If the input list contains duplicate values, the function should return the indices of the first two occurrences that satisfy the condition. Provide the implementation.

CODE EDITOR

 solution.py

Run Code

For Two Sum, implement the function in Python

```
def two_sum (nums, target):  
  
1  def two_sum(nums, target):  
2      seen = {}  
3      for i, num in enumerate(nums):  
4          complement = target - num  
5          if complement in seen:  
6              return [seen[complement], i]  
7          if num not in seen:  
8              seen[num] = i
```

TEST RESULTS

Click Run Code To Test Your Code.

Navigate: Use Prev to return to the previous question and Next to move forward.

Submit: When the assessment is complete, select Submit to submit the assessment.

PREV

3 / 3

NEXT

SUBMIT